

Louis (Yiven) Zhu

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Profile

I study the science of AI evaluation, asking what benchmark scores measure, when they predict performance beyond the test, and when a decision should rely on them. I work quantitatively, with psychometric models (factor analysis, item response theory, reliability), statistical learning (nested cross-validation, held-out prediction) and economics (price indices, panel data). In 2026 I released five sole-authored preprints, on benchmark structure and repair history, LLM-judge reliability, quality-adjusted prices for AI inference and undisclosed change in frontier safety frameworks; two are accepted at NeurIPS 2026 workshops, one is under review at ICLR 2027, and the analysis layer behind three of them is published as the benchprobe library. Ahead of doctoral study from 2027, I am asking whether evaluation results predict how systems perform once deployed, particularly in interactive and agentic settings.

Education

University of Oxford, Oxford Internet Institute (OII)

MSc Candidate, St Antony's College

Oxford, UK

2026 – 2027

University College London, Faculty of Mathematical and Physical Sciences

BSc Science and Technology Studies (First Class Honours, **ranked first of cohort across the Department**)

London, UK

2023 – 2026

- Peter Medawar Prize, awarded to the Department's top-ranked graduate (highest overall result across the full three years); STS Best Dissertation Prize.
- Selected modules: Warnings for All (risk, forecasting and evidence under uncertainty; 88.7); Philosophy of Information (80); Political Economy (80); Governing Emerging Technologies (78). Audited: Microeconomics; Advanced Quantitative Methods; Machine Learning for Economics; Economics of Innovation.
- Dissertation: "The Unassembled Validity Argument: Constructing, Repairing and Circulating MMLU, 2020–2026" (87). [PDF](#).

University of London Online Programme (academic direction from LSE)

BSc Economics and Finance (weighted average 92% to date)

Distance learning, Online

2023 – 2026

- Marks to date: Principles of Asset Pricing (99) and Principles of Corporate Finance (98), **each the highest mark of all candidates in the 2026 examinations**; Derivatives and Risk Management (96); Microeconomics (95); Mathematical Statistics (93); Introduction to Finance (93); Introduction to Economics (91); Financial Data Analysis (87); Mathematical Methods (82); Econometrics (81).

Quantitative Preparation

2025 – 2026

- LSE Summer School 2026, ME315 Machine Learning in Practice (A; final project 90, top of cohort); ME324 Artificial Intelligence and Deep Learning (A–; language-model project 99, top of cohort); EC320 Applied Microeconomics and Big Data (audited).
- MITx MicroMasters in Statistics and Data Science (online, in progress; 2 of 4 courses completed, [Machine Learning](#) and [Data Analysis](#)); University of Oxford, Python for Data Science and Machine Learning & AI in Python (10 CATS each, 100/100).

Relevant Research Experience

London School of Economics (LSE), Department of Statistics

Research Project, Benchmark Measurement (Mentors: Dr [Marcos Barreto](#) and Dr [Thomas Robinson](#))

London, UK

Jun 2026 – Present

- Asked what a frontier leaderboard measures. Found **no distinct economic factor** under the pre-specified dimensionality rule, while under linear learners a multi-factor representation of the other benchmarks predicts held-out economic scores better than a mean-score index, an advantage that reverses for tree learners.
- Built the analysis end to end in Python, from a hash-pinned snapshot of 421 model configurations across twelve benchmarks through release-date-adjusted factor analysis to leave-one-benchmark-out prediction under nested cross-validation, re-estimating the factors in every fold.
- Extended an assessed ME315 project into a sole-authored manuscript, now under review at ICLR 2027, with a [reproduction repository](#) and the analysis layer released as benchprobe.

EvalEval Coalition (Hugging Face · University of Edinburgh · EleutherAI)

Core Contributor, Science of Evaluations (Mentors: Dr [Zeeraq Talat](#) and [Robert Scholz](#))

Remote

Jun 2026 – Present

- Contribute to the paper's treatment of validity and the evidence required to support claims from AI evaluation; manuscript in preparation for TMLR.
- Developed a source-to-claim verification workflow for the draft sections, checking quotations and adjudicating discrepancies against the source material.
- Contribute to Every Agent Ever, the coalition's reporting schema for agent evaluation runs (evalvalai.com).

UCL Department of Computer Science

London, UK

Researcher, Human-Agent Collaboration (Mentor: Prof. [Maarten Speekenbrink](#))

Dec 2025 – Jul 2026

- Co-developed a **1.1-million-run** agent-based simulation of how institutional authority and information structures affect group decisions involving AI agents.
- Co-authored a workshop paper for the Workshop on Human-Agent Collaboration at CHI 2026 ([PDF](#)).

ETH Zurich, Center for Law & Economics

Remote

Research Contributor, [Relit](#) citation-audit tool (Prof. [Elliott Ash](#))

Aug 2026 – Present

- Contribute to Relit, an LLM-based tool that audits whether a paper's claims are supported by, and cited to, the economics literature; benchmark construction and validation workstream.

UCL Department of Computer Science · Holistic AI

London, UK

AI Researcher, Adversarial Robustness of LLM/VLM Agents (Mentor: [Zekun Wu](#), Lead in Agentic AI) Nov 2025 – Jul 2026

- Co-author, **Adversarial Filtered CAPTCHA Attacks on LLM/VLM Agents** (UCL Computer Science & Holistic AI), developing adversarial attacks and robustness evaluation for vision-language and tool-using agents.
- Built Python pipelines to generate adversarial visual inputs, automate agent interactions and evaluate behaviour across visual encoders, LLM policies and safety filters; designed robustness metrics and ablation studies measuring attack success, safety bypasses and cross-model transfer.

Selected Research Outputs

Published and accepted work

The Price of Intelligence: A Quality-Adjusted Price Index for AI Services. Louis Yiven Zhu. 2026. Accepted, NeurIPS 2026 EconML Workshop (Economics for Machine Learning). Preprint, arXiv:2608.29843; journal version in preparation for the Review of Income and Wealth. [Paper](#) · [data](#) · [pre-registered audit](#).

A Score Should Travel With Its Repair History. Louis Yiven Zhu. 2026. Position paper, accepted, NeurIPS 2026 Workshop on AI for Meta-Science. Preprint, SocArXiv, doi:10.31235/osf.io/7bg8r_v2; full version in preparation for the ICML 2027 Position Paper Track. [Paper](#).

Neural Evidence and Behavioural Welfare Economics: When Can Neuroeconomics Inform Policy? Louis Yiven Zhu. 2026. Accepted, UCL Journal of Economics, vol. 5, no. 1 (in proof); doi:10.14324/111.444.2755-0877.2257. Earlier version, arXiv:2511.19548. [Article](#) · [preprint](#).

From Advisor to Voting Teammate: Institutional Authority and Information Structures of AI Agents in Bounded-Rational Human Groups. Tian Y., Zhang J., **Zhu L.**, Li Y., Meganathan A., Niyaa N., Kan H., Purevdorj U., Mon R., Kannan L. 2026. Workshop paper, Human-Agent Collaboration Workshop at CHI 2026. [Paper](#).

Preprints and submitted manuscripts

One Capability or Many? Structural and Predictive Tests of Benchmark Validity Disagree about Economic Benchmarks for Frontier AI. Louis Yiven Zhu. 2026. Preprint, arXiv:2608.29420 (v2, September 2026). Under review, ICLR 2027. [Paper](#) · [code](#) · [analysis plan](#) (retrospective deposit).

Three Ways Classical Test Theory Can Mislead About LLM Judges. Louis Yiven Zhu. 2026. Preprint, arXiv:2609.29709 (v2); full version in preparation for NeurIPS 2027 Evaluations and Datasets. [Paper](#) · [code](#).

Silent Revision: Measuring Undisclosed Change in the Safety Frameworks of Frontier AI Developers. Louis Yiven Zhu. 2026. Preprint, arXiv:2609.08789; full version in preparation for ACM FAccT 2027. [Paper](#) · [corpus and code \(Zenodo\)](#).

Who Makes the Future of Work? Measurement, Hidden Labour and the Evidence on Artificial Intelligence. Louis Yiven Zhu and Joanna Octavia. 2026. Preprint, SSRN, doi:10.2139/ssrn.7544498. Under review, International Labour Review; funded by UCL MAPS. [Paper](#).

Automation Exposure and the UK Labour Market: Employment, Pay and the Wage Floor, 2014–2020. Louis Yiven Zhu. 2026. SSRN working paper, v2, doi:10.2139/ssrn.5736503. [Paper](#) · [code](#).

Selected work in progress

The Science of Evaluation: Open Problems and Opportunities for Development of AI Evaluations. EvalEval Coalition; Louis Yiven Zhu, core contributor. In preparation for TMLR; contributes the treatment of validity and evidentiary standards.

The Unassembled Validity Argument: Constructing, Repairing and Circulating MMLU, 2020–2026. Louis Yiven Zhu. 2026. BSc dissertation (87; STS Best Dissertation Prize). Manuscript in preparation; invited for presentation at the [UCL Centre for Responsible Innovation](#). [PDF](#).

Research Software

benchprobe: psychometrics for AI benchmark scores. Louis Yiven Zhu. 2026. Python library, v0.1.0, MIT; Zenodo, doi:10.5281/zenodo.22705351. Reliability, factor structure, predictive validity and item-response scaling; reproduces three published studies from hash-pinned data, 44 acceptance tests, independent cross-family review. [Repository](#) · [DOI](#).

Further Research Experience

UCL Department of Economics

London, UK

Researcher, Automation Risk and Wage Dynamics in the UK (Supervisor: Prof. [Roland Kappe](#)) Apr 2025 – Sept 2025

- Built an occupation-year panel linking ONS automation-risk scores to occupation-level ASHE wage data, and estimated how real wages evolved in occupations scored as high automation risk after 2016.
- Implemented **two-way fixed-effects estimation** in Stata and Python and documented the limits of a descriptive, occupation-level design; SSRN working paper, v2 (2026).

UCL Faculty of Mathematical and Physical Sciences

London, UK

Summer Research Intern & Departmental Associate, AI and the Future of Work (Dr [Joanna Octavia](#)) May 2026 – Sept 2026

- Appointed Departmental Associate to lead the design of a new UCL module (2026/27) on AI, automation, platform labour and the future of work, a **280-hour funded commission, doubled from 140** after the first deliverables, owning the framework and syllabus.
- Built the evidence base end to end, an 80-hour systematic review of 81 sources coded into a structured field map, now the basis of a review article with Dr Octavia under review at the International Labour Review; internship and article funded by the Faculty (MAPS).

Institute of Economic Affairs

London, UK

Research Contributor (with Economist [Julian Jessop](#)) Aug 2025 – Nov 2025

- Analysed the UK “graduate premium” from GLMS and LEO administrative data, building CPIH-deflated earnings series and separating raw from composition-adjusted premia; wrote policy briefs on the evidence for ministers and journalists.

Technical projects: [hansard-char-lm](#), a character-level GRU language model written from scratch in PyTorch; [DoubleML](#), causal ML experiments; [ABM](#), a labour market under automation shocks.

Talks and Service

Invited talk, One Capability or Many?, Leverhulme CFI lab meeting (J. Hernández-Orallo, L. Pacchiardi), Cambridge	29 Oct 2026
Invited Reviewer, NeurIPS 2026 Trust-AI-Eval (TAE), EconML and JUDGE Workshops	2026
Elected MSc Student Representative, Oxford Internet Institute, University of Oxford	2026 – 2027
Fellow, Thinking About Thinking (AI nonprofit)	2026 – 2027
Director of Engagement, AI for Good Oxford; Sponsorship Lead, Oxford Artificial Intelligence Society	2026 – 2027
President, UCL Political Science & Economy Society; Chairman, UCL Investment Society	2023 – 2026

Selected Awards and Prizes

Peter Medawar Prize for the Highest Overall Result across the full three-year BSc (top cumulative result in cohort), UCL STS	Jun 2026
STS Best Dissertation Prize for “The Unassembled Validity Argument”, UCL STS	Jun 2026
University of London Award for Academic Achievement (one of the best performers on the degree, 2026 examinations)	Sept 2026
STS Joan Beauchamp Proctor Prize for Top Overall Performance (highest cumulative marks across all Year 2 STS BSc modules)	Jun 2025
Explore Econ 2025 shortlisted paper , UCL Department of Economics, on mandatory AI risk disclosure as evaluation governance	May 2025

Skills

- Statistical research: psychometric modelling, IRT, factor analysis, cross-validation, panel econometrics; Python, R, Stata.
- Machine learning: PyTorch language models, evaluation pipelines that run and probe models; NumPy, pandas, scikit-learn.
- Research design: quantitative and mixed-methods research, documentary analysis, source verification and data provenance.
- Languages: English (native); Cantonese and Mandarin (fluent); French (conversational).

Interests: Ice hockey ([BUIHA](#)), alpine ski racing, piano and composition, political philosophy, theoretical physics.
